

Bijection

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1 Bijection

1.1 Bijection Problem

Consider $\ln : (0, \infty) \rightarrow \mathbb{R}$. Is it injective? Is it surjective? Is it bijective?

1. Let $f(x) = e^x$ and $g(x) = \ln(x)$

$$(f \circ g) = z$$

$$z = e^{\ln(x)}$$

$$\ln(z) = \ln(x) = t \quad (\text{adding var } t)$$

$$e^t = z = x$$

$$z = x$$

$$(g \circ f) = y$$

$$y = \ln(e^x)$$

$$e^y = e^x$$

$$y = x$$

1.2 Second Attempt

Proove e^x is injective where $(0, \infty) \rightarrow \mathbb{R}$:

$w_2 > w_1$ and $w_2 - w_1 > 0$

$$e^{w_2} = e^{w_1}$$

$$e^{w_2 - w_1} = 1$$

$$w_2 - w_1 = 0$$

$$\implies w_2 = w_1 \quad \text{contradiction}$$

Proove: $\ln(a) = \ln(b)$

$$\ln(a) = t$$

$$a = e^t$$

$$\ln(b) = t$$

$$b = e^t$$

$$a = b$$

Proove $\ln(x) = y$ is surjective:

$$\ln(x) = y$$

$$\implies e^y = x$$

1.3 Questions

1. Proof: Let $a, b \in \mathbb{R}_{>0}$ such that:

$$\log_2(a) = \log_2(b) \quad (1)$$

Raise 2 to the power of each side:

$$2^{\log_2(a)} = 2^{\log_2(b)} \quad (2)$$

Since 2^x and $\log_2(x)$ are inverse functions:

$$a = b \quad (3)$$

since $a = b$, the function $\log_2$ is injective. This is provable without previous proofs.

2. Is $e^x : \mathbb{R} \rightarrow \mathbb{R}$ surjective:

Let $y = e^x$. For any value of y where $y \leq 0$, so 0 or negative. No value of x can make e^x 0 or negative, so the codomain has to be bound to $\mathbb{R}_{>0}$ for e^x to be surjective. Solution:

$$e^x : \mathbb{R} \rightarrow \mathbb{R}_{>0} \quad (1)$$

3. $x^2 : \mathbb{R} \rightarrow \mathbb{R}$ is not injective, as for any x , $-x$ produces the same output. Restricting the domain $\mathbb{R}_{\geq 0}$ fixes this. The function also can't output negatives even without a restricted domain so:

$$x^2 : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0} \quad (1)$$

0 is included because:

$$0^2 = 0 \quad (2)$$

So it maps correctly, and one to one. So for this restricted domain and codomain, x^2 is injective. The inverse of a square is a square root where:

$$\sqrt{x^2} = x \quad (3)$$

And the $\sqrt{x}$ is valid and injective for any $\mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, but no real values exist for negatives.

4. $x^3 : \mathbb{R} \rightarrow \mathbb{R}$ is bijective because there is an output for each input, including negatives, as odd numbered exponentials always have a leftover negative:

$$x^3 = x \cdot x \cdot x \quad (1)$$

sub -1 for x:

$$-1 \cdot -1 \cdot -1 = -1 \quad (2)$$

or -5:

$$-5 \cdot -5 \cdot -5 = -125 \quad (3)$$

This proves the function is surjective, as all positive and negative values map and continue to do so as $x \rightarrow \pm\infty$.

To prove injectivity, let $a^3 = b^3$ where $b \neq 0$

$$\implies \left(\frac{a}{b}\right)^3 = 1 \quad (4)$$

$$\implies \frac{a}{b} = 1 \quad (5)$$

$$a = b \quad (6)$$

Because 0^x :

$$a^3 \implies a = 0 = b \quad (7)$$

Therefore $x^3 : \mathbb{R} \rightarrow \mathbb{R}$ is injective. And as it is also surjective, it is therefore bijective.